

SIGNIFICANCE

The unmet need. If we are fortunate, each of us will reach the point where we, or someone we love, needs help with the ordinary tasks of daily living: bathing, dressing, meals, medications, and moving safely through the home. Most families meet that moment at its worst, after a fall, a diagnosis, or a sudden hospitalization, and are expected to become experts in eldercare overnight.

A recognized need becomes established care only if a family clears three gates (Figure 1). They must **find** the right care inside a fragmented ecosystem of public aid programs, insurance coverage, healthcare services, and long-term services and supports, each with separate eligibility rules and none accountable for whether care begins. They must **afford** it: full-time home care runs \$80,080 a year,² Medicare does not cover custodial home care, and Medicaid, the nation's largest payer of long-term care, pays only after a family has spent down to eligibility. An estimated \$58 billion in assistance that would close part of that gap goes unclaimed every year because families cannot identify or navigate the programs.¹ And someone must be available to **deliver** the care: 63.3 percent of home-care providers declined cases in 2023 that they could not staff.³

Failure at any one gate produces the same outcome. Needs go unmet, function declines, a crisis arrives with an emergency visit or a hospitalization, and discharge returns the older adult home without the support that was missing in the first place. Nearly a third of older adults with difficulty in activities of daily living report going without bathing, meals, or medications in a given month,⁴ and unmet needs of this kind independently predict emergency department use, readmission, nursing home placement, and mortality.^{5,6,7} Families of modest means therefore wait, and decline care, until a hospitalization, when public programs finally step in at the most expensive possible moment and after the most harm has already occurred.

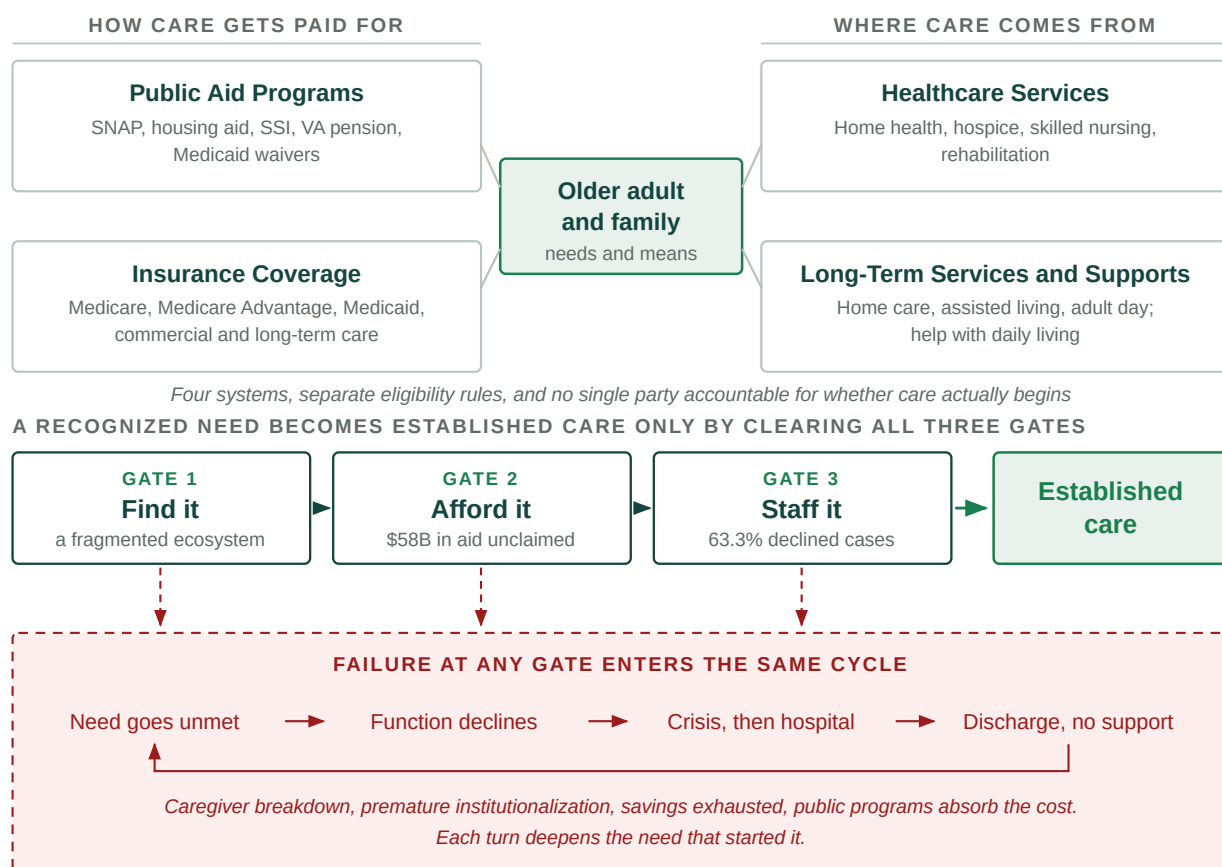


Figure 1. The eldercare ecosystem a family must navigate, the three gates that must all be cleared for a recognized need to become established care, and the cycle that follows failure at any one of them.

What it costs, and who pays it. The gap is widening from both sides. The population over 65 reaches 82 million by 2050,⁸ while the family capacity that has always absorbed most of American eldercare is contracting: the number of potential family caregivers for each adult over 80 falls from more than seven in 2010 to four by 2030 (Figure 2).⁹ The paid workforce that would have to replace it faces 9.7 million direct-care job openings between 2024 and 2034.¹⁰ Demand is rising against two supplies that are falling at once.

The costs of that gap do not disappear. They move, and they grow in transit. Roughly 70 percent of working-age family caregivers hold jobs, and the hours they lose are borne by employers and by their own earnings.¹¹ Families spend down their savings and Medicaid inherits the bill at the institutional rate. The clearest measurement of the inversion comes from CAPABLE, a home-based, function-focused support program whose implementation was associated with roughly \$3,000 in program cost against more than \$20,000 in reduced Medicaid spending per participant, driven by lower inpatient and outpatient use.¹² The same approach reduced disability across multiple randomized trials.¹³

The money to support older adults at home largely exists already. It is simply spent later, on worse outcomes, by whoever is left holding the consequence. Every dollar of that consequence sits on an identifiable balance sheet: a health plan's, a state Medicaid budget's, a health system's, an employer's, or a family's. That is what makes this an addressable market and not only a public health problem.

What establishing care actually requires. Closing the gap is not a matter of better information. A family that knows exactly which program it qualifies for still has to complete the application, gather supporting documents, file, wait, respond to requests, and then find a provider with capacity on the date the benefit starts. Existing help stops before that point. Clinicians and discharge planners hand off. Information services return lists. Referral marketplaces introduce a provider and are paid on placement. None of them is accountable for whether care was established, and none of them can create a caregiver who does not exist.

Two things are therefore required at once: navigation that runs all the way to a confirmed start, and capacity on the other side so that the start is possible. Neither is sufficient alone, which is the central structural claim of this application.

Why navigation alone cannot close the gap. A navigation product that clears the first two gates aggregates family demand and delivers it to a supply side that cannot absorb it. The result is a faster path to the same refusal. This is not a hypothetical constraint: it is the reason nearly two-thirds of home-care providers turned away business in 2023.³

Olera did not select workforce development as an adjacent business opportunity. We encountered it as the binding constraint on our own product working. That is why capacity creation sits inside the product rather than beside it, and why this application proposes to test both halves in the same markets at the same time.

The product and the path to market. CareNavigator is an eldercare AI navigation system built over an expert-curated national database of aid programs and providers, developed across NIA SBIR Phases I through IIB. It screens a household's needs and means, identifies the aid and services that fit, and, with the agent layer entering production in the remaining Phase IIB year, executes the applications and follows each case to a decision. Alongside it, Olera recruits and verifies new caregivers and places them with licensed providers, who hire, train, insure, and supervise them as employees.

Families use CareNavigator at no cost, and no provider pays to be listed or to receive a family connection. Because no fee is charged for a referral, no steering incentive exists and federally reimbursed providers can participate, which pay-per-referral models cannot accommodate. Revenue comes from one optional provider product during this award: **Caregiver Staffing**.

Market segments and customers. The beachhead customer is the healthcare and long-term services provider. Roughly 165,000 such organizations operate in the United States,^{15,16} and they already spend at least \$5 billion a year on marketing and caregiver recruiting, an estimate built entirely from published low-end inputs.¹⁷ They buy from us for an auditable reason. The published median cost of acquiring a caregiver is \$520 per hire, and that figure is for word-of-mouth, the cheapest channel an agency has.¹⁸ Table 1 sets our pricing against that benchmark on the provider's own numbers.

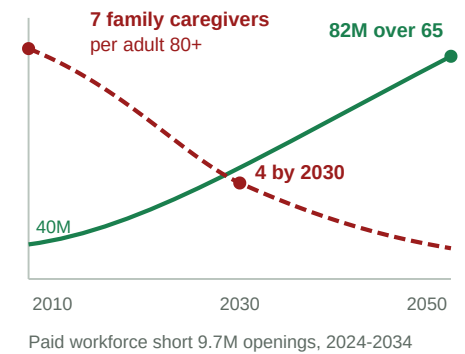


Figure 2. Demand rising while both sources of supply contract.

Agency size	Hires per year at 75% turnover	Current cost at the \$520 median	Olera at \$275/mo, unlimited hiring	Effective cost per hire	Annual saving
15 caregivers	11	\$5,850	\$3,300	\$293	\$2,550
30 caregivers	23	\$11,700	\$3,300	\$147	\$8,400
50 caregivers	38	\$19,500	\$3,300	\$88	\$16,200
100 caregivers	75	\$39,000	\$3,300	\$44	\$35,700

Table 1. What a provider pays today to hire a caregiver, and what Olera charges instead.

We do not ask providers to believe in a new category. We ask them to pay less for something they already buy. Break-even against the benchmark arrives at 6.3 hires a year, which is an agency of roughly eight caregivers, so every larger agency is ahead from the first month. The revenue side is larger still: at 63.3 percent of agencies declining cases they cannot staff, the opportunity cost of an unfilled role dwarfs the subscription price.

A second customer class is emerging and this award evidences rather than sells to it. Medicare now pays directly for care navigation, caregiver support, and respite through the GUIDE model, which began in July 2024.¹⁹ Medicare Advantage plans, Medicaid managed care organizations, health systems and accountable care organizations, self-insured employers, and long-term care insurers each bear part of the cost of unmet need. Aim 3 produces the evidence package these organizations require before contracting; it does not assume they will.

Competitive environment and our advantage. Table 2 places CareNavigator against every category of alternative across the five stages of the path to care. The pattern rather than any single row is the argument: **every alternative solves one stretch of the path and stops at a handoff, and the handoffs are where families are lost.** Human navigators are the strongest competitor and cover four of the five stages; they are also scarce, episodic, and cannot be scaled by hiring, which is the same workforce constraint from a different direction. Staffing channels move workers between employers, so one provider's hire is another's vacancy and the national shortage does not shrink. Referral marketplaces charge placement fees, which determines who appears and has drawn sustained criticism of the incentives it creates.²²

Stage of the path to care	CareNavigator <i>navigation and new workforce</i>	Public information <i>Eldercare Locator, BenefitsCheckUp</i>	General AI assistants <i>ChatGPT, Claude, Gemini</i>	Human navigators <i>social workers, discharge planners</i>	Referral marketplaces <i>A Place for Mom, Caring.com</i>
Assesses needs	✓	✓	✓	✓	✓
Identifies aid and insurance	✓	✓	✓	✓	✗
Matches local providers	✓	✗	✗	✓	✓
Carries the case to established care	✓	✗	✗	✓	✗
Adds the caregivers to staff it	✓	✗	✗	✗	✗

Table 2. Coverage of the path to care, by category of alternative.

Related development efforts in academia and industry. Most of the relevant science is settled and what remains open is operational, which is the reason this is a commercialization award rather than a trial. Research on unmet need, largely from the National Health and Aging Trends Study and Medicare-linked cohorts, establishes that unmet daily-activity needs predict downstream utilization and placement.^{4,5} We treat that as established and measure care established rather than clinical endpoints. Home-based support trials, principally CAPABLE, show that function-focused support reduces disability with savings well above program cost;^{12,13} what no trial addresses is how a family outside a trial finds such support, funds it, and gets it started, and CareNavigator is the front door those programs assume already exists. Work on digital caregiver tools, including four peer-reviewed studies of our own, establishes that such platforms can be built and measured as usable and accepted;^{25,26,27,28} what remains open is whether AI agents can complete applications and carry a case to a decision. Payer-funded navigation is no longer speculative, given the GUIDE model.¹⁹ And research on health-professions admissions establishes that documented patient-care hours are a requirement for

entry,^{23,24} which is the incentive our workforce pathway uses; whether that requirement converts into durable eldercare capacity is what Aim 2 tests.

Hurdles to adoption. The primary hurdle is local density at a cost the market sustains: families, providers, and caregivers must concentrate in the same market, and that market's revenue must cover the cost of bringing them together. Four specific hurdles follow. Families must trust an AI system with care decisions, which we address by drawing every answer from an expert-curated database audited by domain experts and by validating acceptance under IRB approval in Aim 1. Providers are wary of platforms, so they begin at no cost and are asked to pay only after value has been delivered, with conversion and churn measured under real billing in Aim 3. New caregivers must be safe and welcome, which is why licensed providers employ, train, insure, and supervise them, and why Aim 2 measures the pathway end to end. And every market must repeat affordably, which self-service portals and a documented market-entry playbook address, with acquisition cost per market measured in Aims 1 and 2.

INNOVATION

Key Innovation 1: capacity created, not moved. Every staffing channel serving this industry, whether an agency, a job board, or a gig platform, competes for workers already inside the direct-care labor market. One provider's hire is another provider's vacancy, and the national shortage does not shrink by a single worker. Olera adds people who were not in the field (Figure 3).

The first population tested is health-professions students, chosen because admission to their programs requires documented patient-care hours, which makes recruitment structurally inexpensive rather than dependent on wage competition. Health professions confer 263,800 undergraduate degrees a year,²³ and entry into nursing, medicine, and physician assistant training turns on documented direct patient care.²⁴ Our own pilot drew more than 900 applicants. New cohorts arrive every semester, so the pipeline refills rather than depleting.

The innovation is not the student channel, which any competitor could copy. It is the infrastructure underneath it: the machinery that converts a person who has never worked in eldercare into someone a licensed provider will hire, and that carries their verified record of hours, competencies, populations served, supervisor evaluations, and credential status forward across employers. That layer is what makes any new worker pool viable, and its value compounds with every shift worked. Task 2.5 tests a second, non-student pool against the same pathway in two campus-poor markets, converting extensibility from an assertion into a demonstrated result.

Key Innovation 2: execution, not information. The prevailing paradigm is that tools inform and families execute. A family is told which programs it may qualify for and is then left to prepare applications, gather documents, file, wait, respond to requests, and confirm that care began. Those are precisely the steps at which families are lost, and no existing tool is accountable for them.

CareNavigator's agents perform the work and the family decides. **The unit of success changes from information delivered to care established.** That shift is what makes the outcome measurable, what makes it saleable to an organization that bears the cost of unmet need, and what makes the system improve: every completed case becomes supervised training data for the next one. The technical foundation, a multi-agent architecture over a retrieval layer with expert feedback, was built and evaluated under Phase IIB; what this award adds is execution and confirmation.

Key Innovation 3: a county-level record of what actually happens. Eldercare information exists in three layers. The open web holds public directories and program rules, often neither accurate nor current, and it is equally available to any general-purpose AI system. Beneath that sit local program realities and provider truths

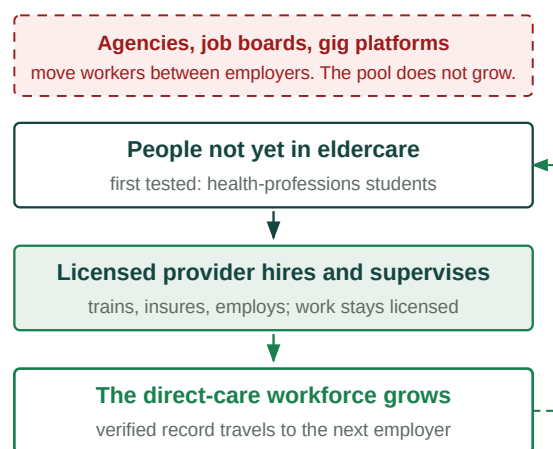


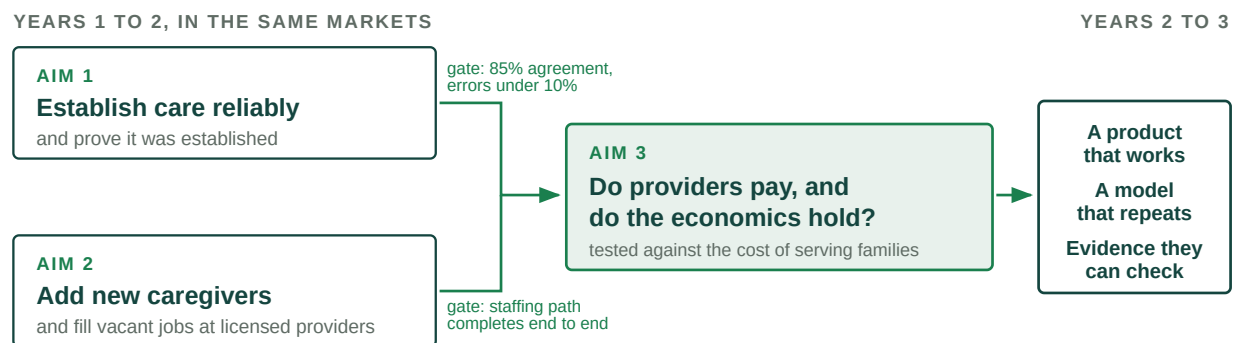
Figure 3. Existing channels move workers between employers. This pathway adds them, and the verified record follows the worker.

that were never published, which we assembled through field work, expert curation, and direct outreach into more than 72,000 records covering all fifty states. Beneath that sits a third layer that can only be learned by operating: what programs actually decide, county by county, where applications stall, whether capacity existed when it was needed, and what care ultimately started.

No public dataset holds the third layer. NHATS is a survey. Claims data record what was billed, not what was needed and refused. **What this project builds is the first longitudinal, county-level administrative record of what older adults need, what they qualify for, where the system fails them, and whether care was established.** Its commercial value is that a competitor who copies every feature begins with an empty database. Its scientific value is that this measurement does not currently exist for researchers or for the state agencies that plan around it, and we commit to publishing from it.

APPROACH

Overall research design and the chain of evidence. Two NIH awards and our own capital have retired five commercial risks, which the CRP Progress Report documents. One remains, and research funding alone cannot retire it: **can a single local market pay for itself, and can we repeat it?** Three aims answer that as one chain of evidence rather than three parallel projects (Figure 4).



Aim 1 without Aim 2 routes family demand at a supply side that cannot absorb it. Aim 3 is meaningless without both.

Figure 4. Three aims, one chain of evidence. Aims 1 and 2 run in the same markets and each carries a gate into Aim 3.

Aim 1 establishes care and proves it was established. Aim 2 supplies the caregivers without whom Aim 1 routes family demand at a supply side that cannot absorb it. Both run in the same markets across Years 1 and 2, because family demand is what makes provider staffing worth buying. Each carries a gate into Aim 3, which asks whether providers pay, whether the unit economics survive real costs, and whether the model repeats. Primary endpoints throughout are operational or economic and come from platform, billing, and employment records.

Three bounded human subjects studies sit inside the aims as Tasks 1.3, 2.3, and 3.3, conducted under Clemson University IRB approval with Dr. Fan as protocol lead. The project is designed to remain outside the NIH definition of a clinical trial: no participant is prospectively assigned to receive or not receive navigation, health-related outcomes appear only as secondary observational measures on a cohort that all receives the service, and the price arms in Aim 3 are assigned to business accounts with purchase as the outcome. Formal determinations will be obtained rather than assumed, including for the streams we expect to be non-human-subjects research, and each determination is recorded in the corresponding task.

Market architecture. The design sets the number of markets, not the reverse (Table 3). **Two anchor markets** open in Year 1 where existing organic traffic and provider density are strongest; they carry the three IRB studies and the deep cost instrumentation. **Eight replication markets** run the documented playbook as written across a two-by-two design on the two axes that plausibly change the economics, workforce source and provider density, with two markets per cell so that variation within a cell can be distinguished from the identity of any single market. Socioeconomic status is measured and reported as a stratification variable rather than made a third axis, because three axes at two levels yields eight cells with one market apiece and no way to separate a market effect from noise.

Ten also satisfies the sample requirement independently. The pricing experiment randomizes accounts that are *offered* a price, and estimating conversion to a 95 percent interval of plus or minus 12 points across three arms

requires roughly 180 offers, which ten markets supply at realistic provider density. **Six of the ten sit as two clusters of three inside single payer footprints**, so that episode volume is concentrated enough to support claims linkage rather than spread thin across unrelated regions. Entry is staged two, four, and four across the three years. At approximately \$30,000 to enter a market, market entry is under eight percent of the requested budget; the binding constraint is operating attention, not capital, and the staging reflects it.

Requirement	What it demands of the design	Markets implied
Price is chosen, not guessed	About 180 accounts offered a price across three adaptive arms under a pre-registered decision rule	4 to 6 at realistic provider density
Unit economics measured, not modeled	Enough paying accounts to estimate acquisition cost, lifetime value, churn, and margin from live billing	4 to 6
The playbook replicates	Two-by-two design on workforce source and provider density, two markets per cell so within-cell variation is separable from market identity	8 replication markets
Depth, instrumentation, and the IRB studies	Markets with existing traffic and provider base, opened first and deeply instrumented	2 anchor markets
Payer evidence	Episode density inside a single payer footprint rather than spread across unrelated regions	Composition, not count: 6 of 10 in two clusters of 3
Operational feasibility and cost	Staged entry a distributed team can run, at about \$30,000 per market to enter	2 in Year 1, 4 in Year 2, 4 in Year 3
Derived design	Two anchor markets plus eight replication markets, staged two, four, and four	10 markets

Table 3. How the market design and its size are derived from what the aims require.

Specific Aim 1: Establish care reliably, and prove it was established. (Years 1 to 2)

Families identify aid and providers on the platform today and are lost while executing applications and again while waiting for decisions. This aim builds the execution and follow-up capability, verifies its output against expert judgment, validates it with family caregivers, and measures what it costs.

Task 1.1: Build the execution and follow-up loops. Three subtasks. *(Task 1.1A) Application execution:* agents assemble and submit applications for the programs identified during screening, with a human review checkpoint before any submission. *(Task 1.1B) Document handling:* the system requests, tracks, and attaches the supporting documentation each program requires, which is the most common point of abandonment in current practice. *(Task 1.1C) Follow-through to decision:* scheduled follow-ups, response handling, and confirmation of the decision and of the date care began.

Task 1.2: Verify output quality against blinded expert review. A panel of licensed social workers and benefits specialists, blinded to whether output was system-generated, independently adjudicates a stratified random sample of cases each quarter for eligibility accuracy, plan appropriateness, and material error. The panel's adjudication is the gold standard against which the system is measured, and disagreement is itself analyzed to improve the retrieval layer.

Task 1.3: Validate usability, trust, and task completion with family caregivers. *(Human subjects research; Clemson University IRB.)* Twenty-five family caregivers recruited from the anchor markets use the integrated system over four weeks. Measures are the System Usability Scale, a validated trust-in-automation instrument, task completion and abandonment points captured from platform telemetry, perception of AI error and the escalation experience, and semi-structured exit interviews. This is an evaluation of an information tool with no clinical intervention and no assignment; a formal determination will be obtained before enrollment.

Task 1.4: Measure cost to acquire and cost to serve from live records. Acquisition cost is measured per family by channel; cost to serve is measured per established case from actual staff time, compute, and support costs, recalculated quarterly so that the trajectory rather than a single figure is the result.

Metrics for Success and Quantitative Criteria (Aim 1):

Measure	Criterion	Source of the measurement
Agreement with the blinded expert panel	≥ 85%	Quarterly stratified case adjudication
Material errors against the expert gold standard	≤ 10%	Same adjudication
Scheduled follow-ups sent on their due date	≥ 95%	Platform telemetry
Usability, System Usability Scale	≥ 72	Task 1.3, IRB study, n = 25
Trust in automation	≥ 5 of 7	Task 1.3, IRB study
Outcome ascertainment at the pre-specified window	≥ 80%	Confirmed care start date
Cost per established case	Falling	Live staff, compute, and support records

Table 4. Aim 1 quantitative success criteria.

Potential problems and alternative strategies. The expert panel is the measurement instrument for Aim 1, so panel disagreement could reflect reviewer variability rather than system error. We will establish inter-rater reliability before any system output is adjudicated and will report it alongside every accuracy figure; cases with panel disagreement are adjudicated by a third reviewer and analyzed separately. Program eligibility rules change during a three-year award, which would degrade accuracy through no fault of the agents; the database maintenance pipeline re-verifies records against current sources on a rolling schedule, and we will report accuracy separately for rules that changed within the measurement window. If agreement stalls below the 85 percent criterion, the alternative is to narrow scope rather than widen it: we will restrict automated execution to the program families where accuracy is highest, route the remainder to human-assisted execution, and report the boundary honestly rather than reporting an average that conceals it.

Specific Aim 2: Bring new caregivers into the workforce. (Years 1 to 3)

This aim tests whether people who were not in the direct-care labor market can be recruited, verified, employed by licensed providers, and retained, and whether providers find the resulting hires valuable. Providers pay nothing for staffing in this aim; paid conversion is tested in Aim 3.

Task 2.1: Verify the recruitment, screening, and verified-record pipeline. Campus and community recruitment channels, eligibility and background screening, training completion, and the construction of the verified experience record that accumulates hours, competencies, populations served, and supervisor evaluations across employers.

Task 2.2: Establish and instrument the placement pathway. Provider onboarding, role matching, offer and hire, and shift confirmation, with each step timestamped so that time to interview, time to hire, and fill rate are measured rather than estimated.

Task 2.3: Validate acceptability, appropriateness, and feasibility with providers and workers. (*Human subjects research; Clemson University IRB.*) Structured assessment using the Acceptability, Appropriateness, and Feasibility of Intervention Measures with provider supervisors and with placed workers, plus semi-structured interviews on safety, supervision, workflow fit, and barriers. Research participation is firewalled from employment decisions, shift allocation, and sales in the protocol, and consent language states this explicitly.

Task 2.4: Measure net-new-worker share and retention. Prior direct-care employment is ascertained at intake, before placement, so that the share of workers new to the field is measured rather than inferred. Retention is reported by cohort at 90 days and at 12 months, and separately for workers who remain with the placing provider and those who move to another employer carrying their record.

Task 2.5: Second worker-pool feasibility pilot. (*Year 3, two campus-poor markets.*) A bounded pilot recruiting career changers and recent retirees through the same pathway, with feasibility endpoints only: whether the pathway can recruit, verify, and place a non-student cohort at comparable cost and retention.

Metrics for Success and Quantitative Criteria (Aim 2):

Measure	Criterion	Source of the measurement
Placed workers with no prior direct-care employment	Reported	Ascertained at intake, before placement
Accounts reaching a first verified paid shift	≥ 70%	Timestamped placement records
Provider acceptability, appropriateness, feasibility	≥ 4.0 of 5	Task 2.3, IRB study
Providers with repeat shift use at 60 days	≥ 50%	Placement records
Worker retention at 90 days and at 12 months	Reported by cohort	Employment and verified-record data
Second-pool feasibility: cost per hire and retention	Comparable	Task 2.5, two campus-poor markets, versus the student cohort

Table 5. Aim 2 quantitative success criteria.

Providers pay Olera nothing for staffing in Aim 2. Paid conversion is tested in Aim 3.

Potential problems and alternative strategies. Retention is the most consequential risk in this application. If workers new to the field leave within 90 days, providers will not pay a second time and the commercial model in Aim 3 weakens regardless of how well Aim 1 performs. Two responses are pre-specified. First, we will report retention two ways: retention with the placing provider, and retention in direct care with any licensed employer, because a worker who moves employers carrying a verified record still represents capacity added to the field even though the original provider lost the hire. Second, if 90-day retention with the placing provider falls below expectations while workforce retention holds, the product emphasis shifts from placement to the portable record itself, which providers value as reduced screening burden. Campus seasonality could create hiring gaps between terms; cohorts will be staggered and the second worker-pool pilot in Task 2.5 will be moved earlier if gaps appear. If providers prove unwilling to employ new entrants at the rate assumed, despite the paid precedent already established, we will shift emphasis to per-diem and supplemental roles, which carry lower onboarding cost and are where declined cases concentrate.

Specific Aim 3: Determine whether providers pay and the economics hold. (Years 2 to 3)

Each offering enters this aim at its gate. Two cohorts are followed: a conversion cohort drawn from Aim 2's free pilots, and a paid-entry cohort that is charged from first contact, so that willingness to pay is not confounded with the experience of having received the service at no cost.

Task 3.1: Set price and packaging under real billing. Offer prices are randomized at the account level, stratified by market and provider size, with three arms anchored on the prices providers have already paid us, approximately \$275 per month and approximately \$150 per hire. The design is adaptive: a pre-registered interim analysis at month 30 drops a dominated arm and reallocates. The decision rule, registered before the first offer, selects the price maximizing expected 12-month revenue per account rather than conversion alone, because a price that converts and then churns is worse than one that does neither. Contamination is monitored, markets are paired non-adjacently, and market-level assignment is pre-specified as a sensitivity analysis.

Task 3.2: Measure unit economics and retention from live billing. Paid conversion within 60 days of offer, revenue per account, cost to serve one paying account from actual billing and support records, retention at 6 and 12 months, churn by survival analysis, customer acquisition cost, payback period, and per-market contribution margin after the cost of serving that market's families.

Task 3.3: Qualitative adoption, perceived value, and price fairness. (*Human subjects research; Clemson University IRB.*) Semi-structured interviews with approximately 30 provider decision-makers spanning converters, non-converters, and churned accounts, examining what was purchased, what value was perceived, how price was judged, and why accounts lapsed.

Task 3.4: Independent rebuild of the economics. An independent financial analyst engaged in Year 3 reconstructs unit economics from raw billing, payroll, and cost records without access to our model, and reports discrepancies. This converts a claimed number into an audited one, which is the form an investor and a reviewer both require.

Task 3.5: Assemble the payer evidence package. Negotiate and execute data-use agreements with at least two risk-bearing organizations; demonstrate that our episode records link to their claims or encounter data at a pre-specified match rate; deliver intermediate outcomes with face validity, namely aid dollars secured, needs

closed, time to established care, and share of episodes ending in established care; and deliver an actuarial cost model built on published effect sizes with sensitivity analysis and stated assumptions.

Metrics for Success and Quantitative Criteria (Aim 3):

Measure	Criterion	Source of the measurement
Paid conversion within 60 days, by offered price	Estimated to ± 12 pts	Randomized offers under real billing
Cost to serve one paying account	Measured	Billing, support, and payroll records
Lifetime value against acquisition cost, 12 months	$\geq 3 : 1$	Independent rebuild, Task 3.4
Payback period on acquisition cost	< 12 months	Same
Per-market contribution margin after cost of serving families	Positive	Same
Executed data-use agreements with risk-bearing organizations	≥ 2	Task 3.5
Claims linkage at the pre-specified match rate	Demonstrated	Task 3.5

Table 6. Aim 3 quantitative success criteria.

Potential problems, alternative strategies, and the stop rule. Account-level price assignment invites contamination, because providers in a market talk to one another. Individualized quotes are normal in business software, and the pre-registered mitigations are a contamination monitor built into the offer workflow, non-adjacent market pairing, and market-level assignment specified in advance as a sensitivity analysis. Conversion could prove too low at every price for the arms to be distinguishable; the adaptive design reallocates away from dominated arms at the interim analysis, and if all arms fall below 20 percent, that is itself the finding rather than an inconvenience. Negotiating data-use agreements with risk-bearing organizations is outside our control; we will pursue at least two independent counterparties in parallel beginning in Year 1, and if none executes by month 30 we will deliver the cost model against a de-identified internal cohort and state the linkage limitation plainly rather than deferring the deliverable.

A stated stop rule. If, at the month-24 decision point, fewer than 40 percent of placed workers remain in direct care at 90 days *and* paid conversion is below 20 percent at every offered price, we will report the provider-funded model as disconfirmed, halt market expansion at wave one, and redirect the remaining effort to completing the payer evidence package and publishing the negative result. A commercialization program should fund projects that are willing to find out they are wrong, and we would rather report that outcome than spend Year 3 defending it.

What this award establishes, and what it prepares. A risk-bearing organization requires six things before it will contract: a measured case with verified establishment, a way to match its members to our cases, demonstrated linkage to its own claims records, intermediate outcomes it values, a defensible cost model, and proof of reduced hospital use and institutional placement. This award delivers the first five. **It does not deliver the sixth, and we will not claim it does.** Establishing that navigation causally reduces utilization requires a design this project deliberately does not run, because running it would make the award a clinical trial and would answer a question no payer will fund us to ask before the first five items exist. The refusal is what makes the first five credible, and completing them is what makes the sixth fundable by someone other than NIH.

Technical assistance and project oversight. Three external providers carry defined deliverables. **Clemson University**, through Dr. Fan as Co-Investigator at 25 percent effort, designs and executes the three human subjects studies and holds the protocols; integration runs through joint quarterly protocol review and shared data-management procedures specified in the subaward. **An independent financial analyst**, engaged in Year 3, performs Task 3.4. **Actuarial consultation** supports the cost model in Task 3.5. The Principal Investigator holds integration authority and all go/no-go decisions, supported by a weekly cross-aim operating review and a quarterly milestone review against the criteria in Tables 4 through 6. Consultant scopes, deliverables, and reporting cadence are specified in the Project Management Plan.

Regulatory plan. No federal premarket pathway applies. CareNavigator is navigational and administrative software that does not diagnose, treat, or make clinical decisions, and therefore is not a medical device; no premarket authorization gates commercialization and nothing in this plan waits on a federal approval. Four regimes do govern and are addressed operationally. Family data is handled under HIPAA-aligned safeguards.

Because no fee is charged for a referral, no steering incentive exists, which is also what allows federally reimbursed providers to participate. Employment, training, insurance, and supervision of every placed worker rest with the licensed provider, not with Olera. And no payer record is joined without an executed data-use agreement and an IRB determination, which Task 3.5 sequences accordingly.

Timetable, decision points, and the state at award end. Figure 5 gives the quarter-by-quarter schedule, the sequencing among the aims, and three formal decision points. At month 12, the execution loops must be verified and the staffing pipeline running before wave one opens. At month 24, the Aim 1 and Aim 2 gates must be met and the stop rule evaluated before wave two opens. At award end, the independent rebuild and the payer package are delivered.

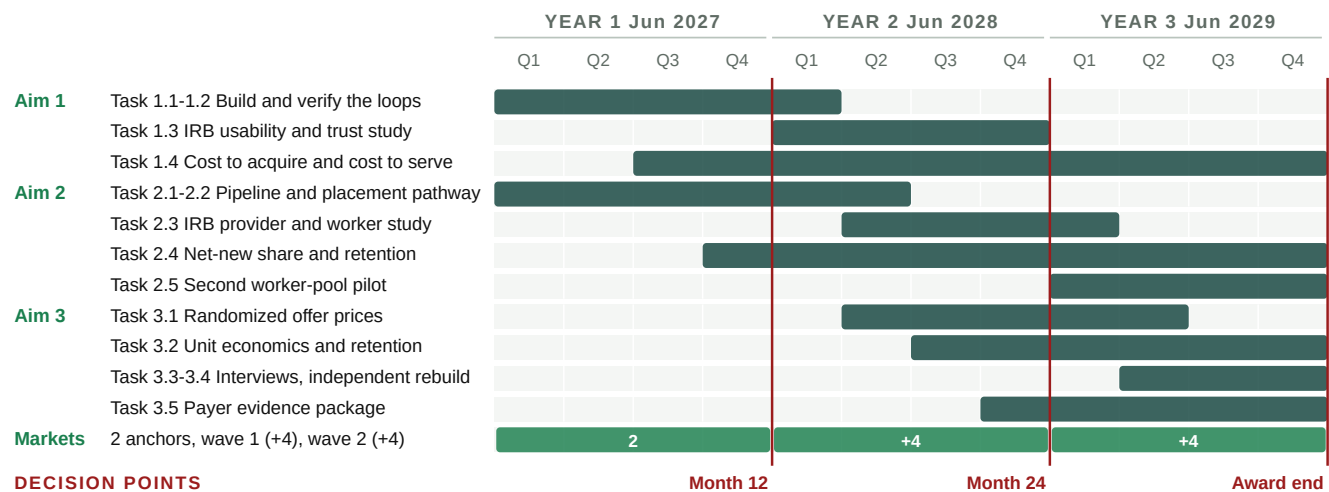


Figure 5. Three-year timetable, showing the sequencing among the aims, staged market entry, and the three formal decision points.

Successful completion leaves Olera holding four things that do not exist today: a product verified against blinded expert review, a price chosen by experiment rather than assumption, unit economics an outside analyst rebuilt from records, and a market-entry playbook run as written across four market types with its variance reported. Together with signed data-use agreements, demonstrated claims linkage, and a delivered cost model, that is the evidence a private investor requires to fund national expansion and the evidence an institutional payer requires to begin its own validation.

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Development status of the technology. CareNavigator operates in production today. Families screen their needs and means, receive matched aid programs and providers drawn from an expert-curated national database of more than 72,000 records covering all fifty states, and receive AI-drafted, expert-approved guidance. The multi-agent navigation layer, developed and evaluated under Phase IIB using retrieval-augmented generation, parameter-efficient fine-tuning, and reinforcement learning from expert feedback, is in production integration during the remaining Phase IIB year. Execution and follow-up, the capability this application builds, do not exist in the product today and are not funded by any existing award.

What Phase I through IIB established. The technology was developed across NIA SBIR Phase I/II Fast-Track and Phase IIB awards (1R44AG074116), which received impact scores of 20 and 25 respectively. Four peer-reviewed studies evaluated the platform with family caregivers: usability scored 4.57 of 5 on the Mobile Application Rating Scale, and technology acceptance scored 5.83 of 7 after four weeks of independent use among 65 caregivers, with every item above 5.0 and higher use frequency associated with higher acceptance.^{25,26,27,28} The multi-agent version scored 5.73 of 7 among 31 dementia family caregivers in a study now in preparation.²⁹ The Phase IIB review assessed the platform's commercial potential as extremely high, citing its ability to help caregivers access eligible funding.

Commercialization progress, including work beyond the Phase II scope. Table 7 records what each activity retired and which funding produced it. Two of the five risks were retired outside the Phase II scope, using I-Corps support and the company's own capital, which is the record of commercialization initiative this program is designed to reward.

Risk	Retired by	Evidence
Technical Could the data be assembled?	NIA Phase I to IIB	An expert-curated national database of more than 72,000 aid programs and providers across all fifty states, and a multi-agent eldercare AI that runs on it, now entering production integration.
User Would families accept it?	NIA Phase I to IIB	Four peer-reviewed evaluations with family caregivers: usability 4.57 of 5; technology acceptance 5.83 of 7 after four weeks of independent use (n = 65); multi-agent version 5.73 of 7 (n = 31, in preparation).
Demand Would families come, and at what cost?	Beyond Phase II scope; company funds	Organic traffic grew from roughly 50 visits a day in 2023 to more than 500 today, about 15,500 a month, arriving through search from nearly every county in the country at near-zero acquisition cost.
Supply Would providers participate?	I-Corps and company funds	More than 300 customer-discovery conversations with owners and operators through NIH and NSF I-Corps. More than 700 providers have since claimed their listing at no charge, with roughly 150 more each month.
Workforce and willingness to pay Could we recruit caregivers, and would anyone pay?	Beyond Phase II scope; company funds	A student caregiver pilot drew more than 900 applicants and placed more than 20 into provider jobs. Four providers trialed the staffing product and three paid, at roughly \$50 to \$275 per placement and \$275 per month, across multiple semesters. Delivery was manual, which capped scale; this award builds the infrastructure that removes the cap.
The risk that remains Can one local market pay for itself, and can we repeat it?	Not yet retired	Measurable only with real customers paying real prices over a retention window long enough to observe churn. This is the uncertainty Aims 1, 2, and 3 are designed to remove.

Table 7. Commercialization risks retired to date, the funding that retired each, and the risk that remains.

Ongoing Phase IIB work this award does not re-fund. The remaining Phase IIB year completes production integration of the agent system and continues database expansion. This application begins where that work reasonably ends and duplicates none of it. No budget line in this proposal supports agent development already funded under Phase IIB; the engineering requested here builds execution, follow-up, confirmation, and the workforce infrastructure, none of which is in scope for the existing award.

The team that will execute this work. Dr. Falohun has served as Principal Investigator on both prior NIA awards, which were scored at 20 and 25 and delivered against their milestones. Dr. DuBose, a physician with an MBA, leads product and market operations and built the organic acquisition channel and the provider pipeline described above. Dr. Fan, at Clemson University, serves as Co-Investigator at 25 percent effort and leads all human subjects protocols; she is a co-author on the platform evaluation record cited here. Dr. Marcia Ory of Texas A&M University, whose NIH research career spans aging services and implementation, advises on study design and dissemination and is likewise a co-author on that record. David Qu advises on commercialization, go-to-market, and investor readiness, and has reviewed and pressure tested the commercial plan in this application.

Olera operates as a fully distributed company: two full-time and two part-time employees, dedicated engineering and marketing support, a two-person care navigation call center, and contract staff engaged as work requires, scaling with this award into market operations and workforce operations roles. Distributed operation is not a limitation for this project but the reason a ten-market design is feasible for a company of this size, because no market requires a local office. Staged entry, two markets in Year 1 and four in each subsequent year, is matched to that operating model and gated at each wave.

The risk that remains. Nine years of federal and company investment retired every commercial risk that could be retired without customers. What none of it purchased is proof that a local market pays for itself and that the result repeats, because that cannot be purchased. It has to be measured in real markets, with real money changing hands, over a period long enough to observe retention and churn. **That is the uncertainty the three aims are designed to remove, and it is the reason this application exists.**

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